

Edutech Solution

AI & ML INTERNSHIP

Task 8: Simple & Multiple Linear Regression

Tools:

- Python (Scikit-learn, Pandas, Matplotlib)

Dataset:

- "HousePrices Dataset" or "Advertising Dataset"

Hints / Mini Guide:

1. Split the dataset into features (X) and target (y).
2. Perform a train-test split (e.g., 80/20).
3. Implement Simple Linear Regression using a single feature.
4. Implement Multiple Linear Regression using multiple features.
5. Visualize the regression line for the simple model.
6. Predict values on the test set.
7. Evaluate using Mean Squared Error (MSE) and R-squared score.

Deliverables:

- Jupyter Notebook with regression implementation and evaluation
- Visualizations of the regression line and error analysis

Final Outcome:

Intern understands the mathematical and practical application of predictive modeling.

Interview Questions Related To Above Task:

- What are the assumptions of Linear Regression?
- Explain the difference between R-squared and Adjusted R-squared.
- What is the cost function used in Linear Regression?
- What is the significance of the intercept and coefficients?
- How do you handle multicollinearity in Multiple Linear Regression?

Submission Guidelines

- **Time Window:**

You can complete the task any time between 10:00AM to 10:00PM on the given day. Submission closes at 10:00 PM.

- **Self-Research Allowed:**

You are free to explore Google, or refer to tutorials to understand concepts and complete the task effectively.

- **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- **GitHub Submission:**

Create a new GitHub repository for each task. Add everything used - code, datasets, screenshots, and a README.md.

- **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link provided.

URL : https://docs.google.com/forms/d/e/1FAIpQLSf_3RtphYvxqIgBmtDQLoiBBN6lSKrYX25WdnzXsSA38_ICYg/viewform?usp=dialog

Best of Luck